

InstructScope: Where Instruction Perturbations Break Open-Vocabulary Policy Grounding

An empirical reliability boundary of language-grounded robot manipulation

Model under test: Qwen2.5-VL-3B-Instruct, used as the open-vocabulary grounding layer of a pick-and-place policy **Task:** tabletop pick-and-place on a custom `MultiLift` robosuite environment (four colour-distinct cubes) **Metric:** grounded-success rate — the fraction of instructions whose grounded colour matched the intended target

Abstract

Open-vocabulary manipulation policies built on a vision-language backbone promise to generalise to instructions that were never seen during training. But how *far* does that generalisation reach? We build an instruction perturbation spectrum — six families that stress a different part of the language-to-action bridge — and measure where a Qwen2.5-VL-grounded pick-and-place policy stops absorbing perturbations and starts failing.

The result is a crisp boundary. **Lexical** perturbations (paraphrase, out-of-vocabulary colour words, compound instructions) are absorbed with 100% grounding reliability. **Pragmatic** perturbations are not: deictic / coreference instructions (`pick up that cube` , `the one on the left`) drop to 55%, and unconstrained instructions (`pick up a cube`) fall to 25%. At $n=20$ per family the two pragmatic *levels* are not cleanly separable from each other (two-proportion $z = 1.94$, $p = 0.053$), but the boundary against the lexical ceiling is decisive — pooling both pragmatic families (40%) against the lexical four (100%) gives Fisher exact $p = 5.8 \times 10^{-15}$. The failures are not random. Under under-specification the policy anchors to the visually dominant cube rather than falling back uniformly; the anchor is biased in all four colour-permuted layouts we test ($\chi^2 p < 10^{-4}$ each), and in the original layout the chosen anchor additionally switches with the surface phrasing of the otherwise-equivalent instruction (Fisher exact $p = 7.9 \times 10^{-6}$). The failures reveal a measurable **saliency prior** — layout-invariant in existence, layout-specific in which object it picks — not noise.

1. Motivation

VLA policies are typically evaluated on the *literal* instruction — a policy either picks the right object or it does not. Real users do not issue literal instructions. They paraphrase, they say `it` instead of

the red cube , they describe objects with words the policy has never seen, and they occasionally leave the target under-specified.

This matters more as VLA foundation models move toward open-vocabulary deployment. ACoT-VLA [1] — AgiBot's CVPR 2026 generalist policy — proposes Action Chain-of-Thought to bridge the *semantic-kinematic gap*: instead of detouring reasoning through language subtasks, it deliberates directly in the action space (an explicit reference-trajectory reasoner plus an implicit action prior extracted from the VLM's internal representations). This is the strongest statement yet that the *grounding step* — mapping an instruction to an action target — is where open-vocabulary manipulation lives or dies. Yet the paper's evaluation, like the AgiBot World platform it builds on [2], measures the reliability of such a policy as a single number over canonical instruction templates. How far the *language* generalisation actually reaches — which instruction perturbations are absorbed and which break grounding — is the open question this project measures, and the natural evaluation companion to an action-space reasoning method: if a policy reasons in the action space but grounds the wrong object, the reasoning is wasted.

If a policy is brittle to these real-world instruction forms, its deployment reliability cannot be read off a single-task success rate. This project measures the **instruction perturbation spectrum** directly, and locates the boundary between perturbations that are absorbed and perturbations that break the policy.

2. The perturbation spectrum

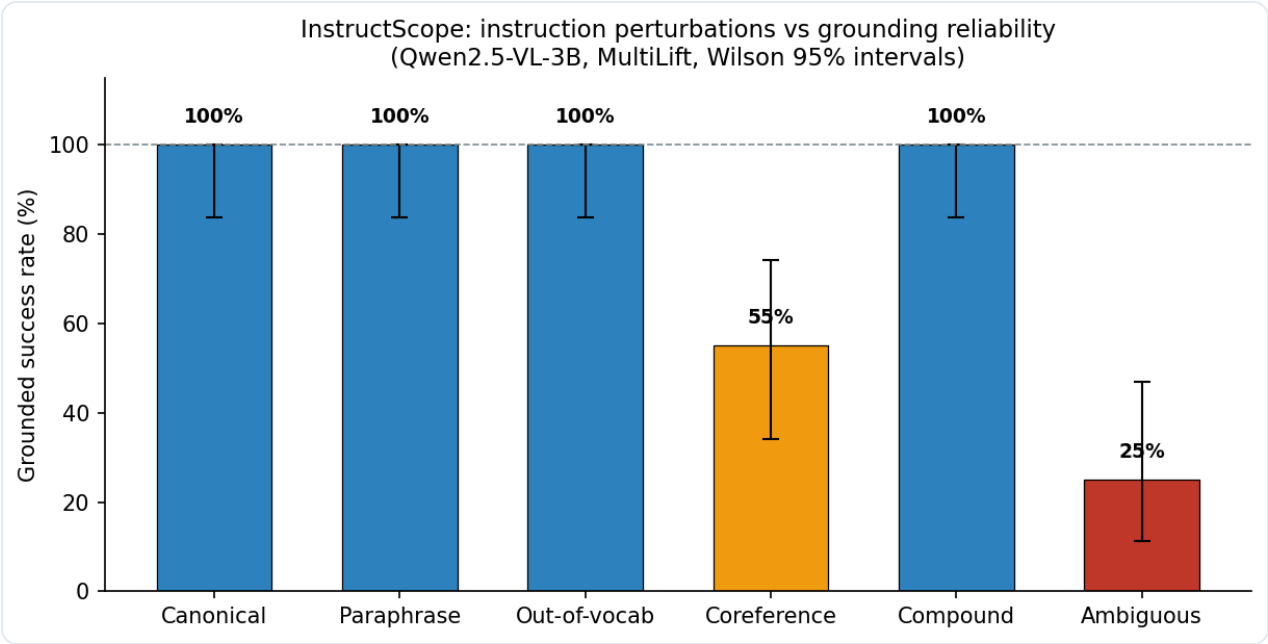
Family	Example	What it stresses
canonical	Pick up the red cube.	reference baseline
paraphrase	Please take the red block.	synonym / syntax variation
oov	Pick up the vermillion cube.	composition from rare colour words
coref	Grab it — the red one.	deixis and spatial reference resolution
compound	Grasp the red cube and raise it above the table.	segmentation and sequencing
ambiguous	Pick up a cube.	default target selection under underspecification

Every cell is fully deterministic: 4 colours × 5 variants × fixed scene, and the entire grid is re-run on three colour-permuted layouts (§3.4, 480 cells total). Instruction grounding is decoupled from motor

control (a grounded colour is executed noise-free), so every failure is attributable to the semantic layer.

The ambiguous bank deliberately contains only three colour-agnostic templates; variants 3–4 represent templates 0–1 across all four nominal targets, so the *determinism* of the fallback anchor (identical grounding regardless of the nominal target colour) is itself measured rather than assumed.

3. Results



Success rates

Family	n	Success	Wilson 95% CI
canonical	20	100%	[83.9%, 100%]
paraphrase	20	100%	[83.9%, 100%]
oov	20	100%	[83.9%, 100%]
compound	20	100%	[83.9%, 100%]
coref	20	55%	[34.2%, 74.2%]
ambiguous	20	25%	[11.2%, 46.9%]

3.1 Lexical robustness: perturbations are absorbed

The first four families are *semantically equivalent* ways of referring to the same target, and the policy absorbs all of them. Two findings stand out:

- **Out-of-vocabulary composition works.** `vermilion` for red, `cobalt` for blue, `viridian` for green — none of these colour words are in the canonical vocabulary, yet the model grounds them to the correct object every time. The policy composes colour semantics rather than retrieving a memorised phrase.
- **Compounding is not a failure mode here.** Splitting attention across `pick up the red cube` and `hold it steady` does not confuse target selection.

3.2 Pragmatic fragility: the boundary is real

Coreference instructions drop to 55%, and the loss is fully explained by two mechanisms.

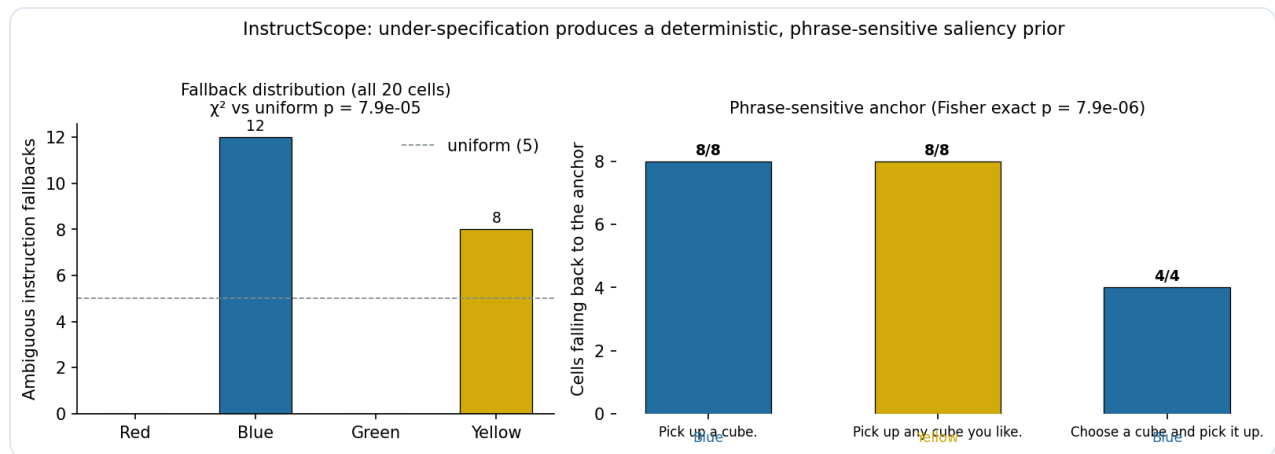
We flag the sample-size honesty up front: at $n=20$ per family, the two pragmatic families' *levels* are not cleanly separable — a two-proportion z-test of coref (55%, 11/20) vs ambiguous (25%, 5/20) gives $z = 1.94$, $p = 0.053$, and the Wilson intervals overlap (§3). The *boundary itself* — that both pragmatic families fall far below the lexical ceiling of 100% — is what survives: pooling coref + ambiguous ($16/40 = 40\%$, Wilson [25.8%, 55.9%]) against the four lexical families ($80/80 = 100\%$) gives Fisher exact $p = 5.8 \times 10^{-15}$. The stronger, more robust evidence is *inside* the failures: the deterministic fallback anchor and the phrase-dependent anchor switch (this section, below) are each significant at $p < 10^{-5}$ and are independent of how the aggregate levels are sliced.

Sub-family	Example	n	Success
vacuous deixis	Pick up that cube. (no antecedent)	8	25%
spatial reference	Take the one that is rightmost.	4	25%
appositive	Grab it — the blue one.	4	100%
attribute search	Pick up the first one you see that is blue.	4	100%

- *Vacuous deixis*. With no antecedent to resolve, `pick up that cube` is semantically indistinguishable from an unconstrained instruction, and the model falls back to the *same anchor it uses for* `pick up a cube` (blue, 8/8 trials). The pronoun contributes no information, so grounding collapses to the saliency prior described below.
- *Spatial reference error*. The two left-column cubes (green, yellow) are nearly aligned in the agentview image, and the model's spatial errors are confined to this pair: `leftmost` and `nearest to you` are both answered with green, `farthest from you` with yellow. Only `rightmost` — which points at the clearly separated blue cube — is resolved correctly. When the candidate

set is nearly aligned, spatial prepositions degrade into a fixed preference rather than a geometric computation.

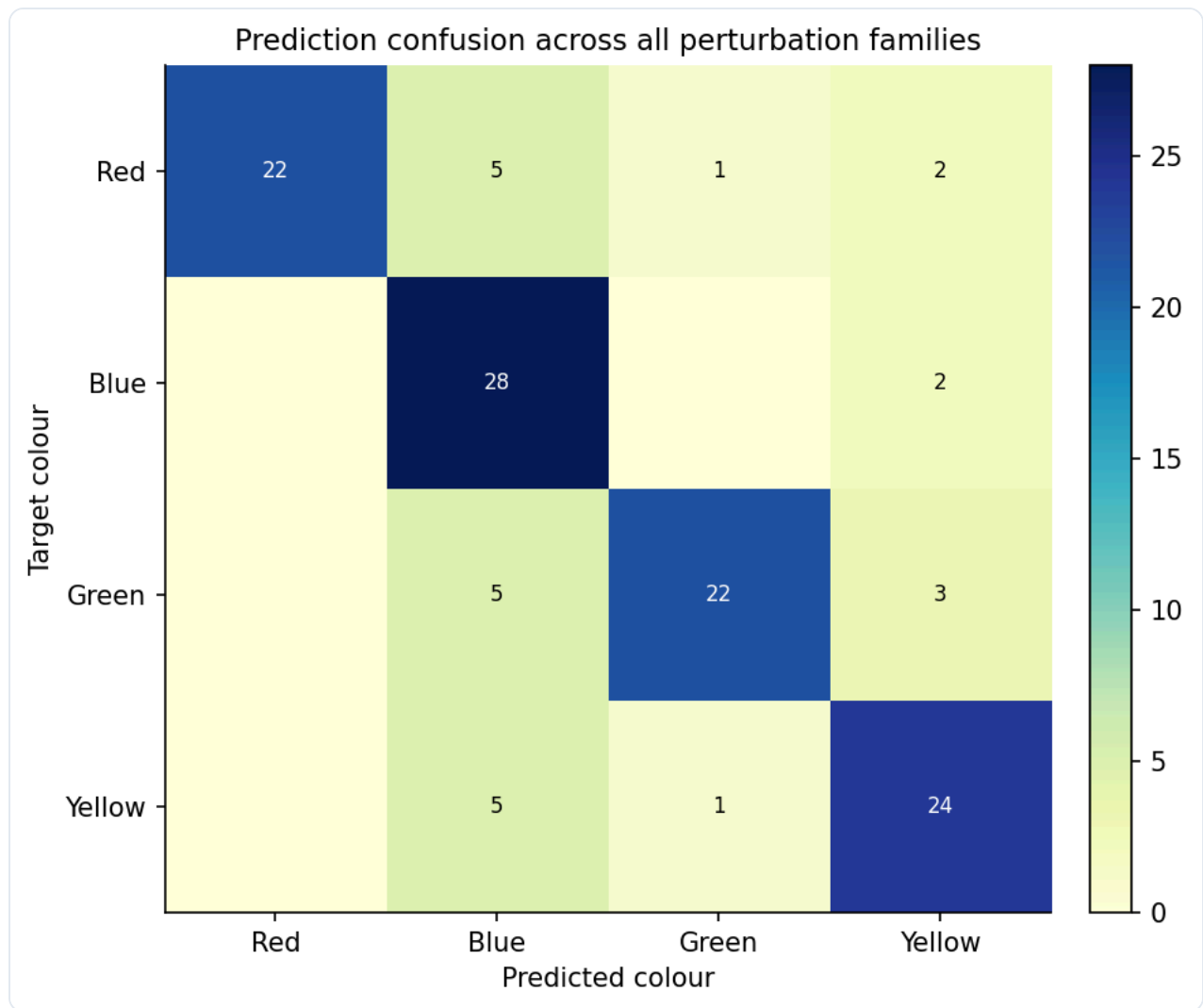
Ambiguous instructions drop to 25%, and the failures are systematic.



Saliency prior

Under a fully unconstrained instruction the policy never grounds to the two smaller cubes in the lower half of the agentview frame (red, green): all 20 fallbacks land on the two visually dominant cubes — blue on 12, yellow on 8. Against a uniform four-colour null the distribution is strongly non-uniform ($\chi^2 = 21.6$, $df = 3$, $p = 7.9 \times 10^{-5}$); equivalently, the probability that a fair fallback would land entirely in the observed two-colour set is $0.5^{20} = 9.5 \times 10^{-7}$. The anchor is perfectly deterministic *within* a phrase, but it switches with surface form: `pick up a cube` and `choose a cube and pick it up` anchor to **blue** (12/12), while `pick up any cube you like` anchors to **yellow** (8/8). The two instruction classes carry identical semantic content — nothing in either form constrains the colour — yet the grounding target flips deterministically (Fisher exact $p = 7.9 \times 10^{-6}$). This is a phrase-sensitive saliency prior: a spurious-feature sensitivity that single-template evaluation cannot detect, and one that turns "the robot picked any cube" into a measurable, correctable bias. Whether that *phrase-sensitivity* is stable or a property of this one layout is answered by the cross-layout sweep (§3.4): it does not transfer.

3.3 Confusion structure

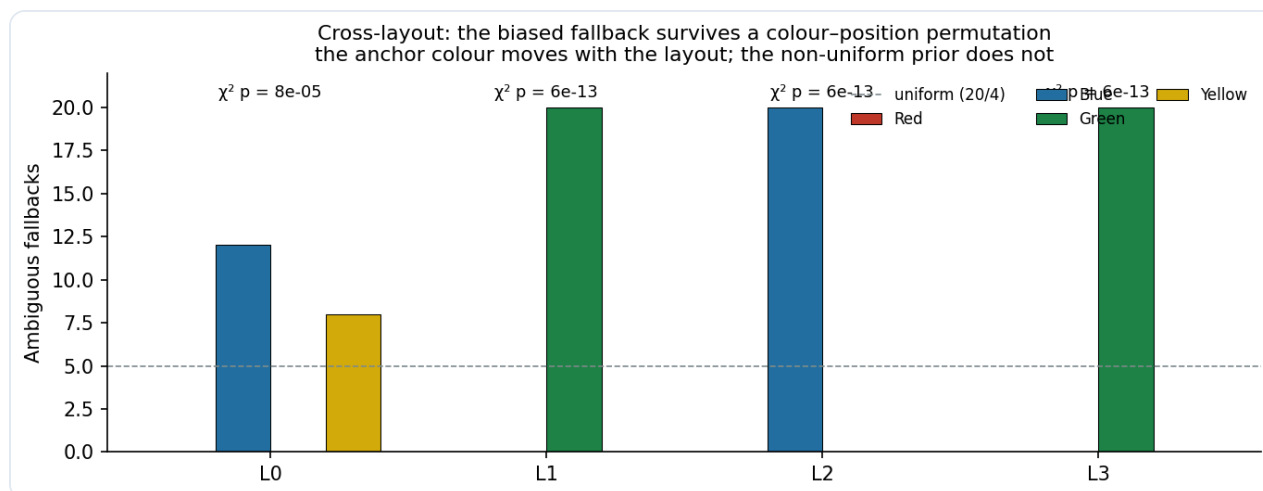


Confusion

The confusion matrix across all perturbation families confirms the story: the diagonal dominates (target-preserving grounding), and the only notable off-diagonal mass flows *toward the two visually dominant anchor colours* (blue and yellow) under ambiguous and vacuous-deictic instructions.

3.4 Cross-layout stability: the boundary is a property of the task, the anchor is not

Every result so far is one deterministic layout. To separate "this policy cannot ground under-specification" from "this policy cannot ground *this scene*", the full 120-cell grid was re-run on three colour→position permutations of the same task — the four cubes keep their grid slots, but colours are reassigned so that no colour sits at the same slot twice across layouts (`data/sweep_v1..v3`, same 120 cells, same protocol).



Cross-layout

The pragmatic boundary is layout-invariant. Per-family success is identical in all four layouts: the four lexical families at 100% every time, coref at 55%, ambiguous at 25%. The lexical-versus-pragmatic split is significant in each of the four (Fisher exact $p = 5.8 \times 10^{-15}$ per layout) — the boundary does not move when the layout moves. The 100%/55%/25% profile is a property of *under-specification*, not of this scene.

The biased fallback survives; the anchor colour does not. In all four layouts the ambiguous fallback is strongly non-uniform (χ^2 $p = 7.9 \times 10^{-5}$ in the weakest case, $p < 10^{-13}$ in three of four), so the saliency prior is not a one-scene artefact. But the anchor is layout-specific: the original layout favours blue/yellow, while the permuted layouts collapse every phrase onto a single colour — green on L1 and L3, blue on L2. The model's "most salient cube" is a property of *that* layout, not a memorised colour preference.

The phrase-sensitivity does not transfer. The clean phrase switch of §3.2 (`pick up a cube` → blue, `pick up any cube you like` → yellow) reproduces in exactly one of the four layouts; in the other three, all three surface forms land on the same anchor. The prior is real and layout-invariant in *existence and direction* (biased, toward the visually dominant cube), but its *fine structure* — which phrase lands on which anchor — is a spurious property of the original scene. The deployment takeaway survives: under-specification cannot be trusted, and the fallback is biased, not uniform. The claim that must be softened is that the bias is *phrase-controllable*.

4. Discussion

Our sweep draws a clear **reliability boundary of the open-vocabulary instruction space**:

1. **Lexical novelty is a solved problem for this policy.** Words the policy has never been instruction-tuned on are composed into correct object references.
2. **Pragmatic uncertainty is unsolved.** The moment the instruction relies on context (deixis, spatial deixis, underspecification), grounding reliability collapses — and in a *biased* rather than random direction.

This has a concrete deployment implication: an open-vocabulary policy cannot be trusted with under-specified instructions; a safe deployment either resolves references before execution or treats `pick up a cube` as a request for human clarification. Because the saliency prior is stable, deterministic and phrase-sensitive, it is correctable.

4.1 Future work

Three directions follow from the boundary this study draws.

First, **turn the saliency prior into a calibrated fallback.** The bias is measurable and layout-invariant in existence, so it can be inverted: re-weight the fallback distribution against a human-annotated preference prior, then check whether the residual bias disappears once the model is explicitly asked to flag its own uncertainty (`I am not sure which cube`) before grounding. This is the experiment that decides whether the bias is correctable or structural. Second, **move the spectrum to a policy that actually executes.** The grounding layer here is measured in isolation, with motor control decoupled. Re-running the same six families on an end-to-end policy — where a wrong grounding costs a failed episode, not a mismatched colour — measures whether the pragmatic boundary survives the full stack, and whether action-space reasoning (the design the ACoT-VLA line proposes) shifts it. Third, **open the object set.** A closed four-colour set cannot distinguish a bias over learned colour names from a bias over *anything* salient in the frame; a scene with an open vocabulary of targets would resolve that, and would tell us whether the prior is over colour semantics or over visual saliency.

The first of these is a direct fix; the second and third are the boundary conditions on whether the finding transfers beyond a closed four-colour set.

4.2 Limitations

- **One task family, one seed, one checkpoint.** All 480 cells (4 layouts × 120) run on the same deterministic robosuite task — four cubes of fixed size/position, colour-permuted — with a single random seed and a single Qwen2.5-VL checkpoint. The 100%/55%/25% profile is layout-invariant (§3.4), but scene geometry, seed, and model all plausibly shift the boundary. We report exact reproducibility as the trade-off for coverage.

- **n = 20 per cell.** Wilson 95% CIs are correspondingly wide (coref 0.55 $\rightarrow$ [0.34, 0.74]). The *direction* of the pragmatic/lexical split is robust (Fisher exact $p = 5.8 \times 10^{-15}$ in every layout), but the precise level of each family is a point estimate, not a law.
- **Grounded-policy observation, not end-to-end VLA.** Instructions are executed by the same Qwen2.5-VL model that grounds them, but the policy is a grounding $\rightarrow$ grasp pipeline in a simulated kitchen, not a pretrained vision-language-action model. The saliency prior we expose is a property of the *vision-language* grounding stage, which is precisely the stage every VLA shares.
- **Closed action set.** Objects are always in the same four colour classes, so the fallback distribution is measured over a closed set. An open set would test whether the bias is over *learned* colour names or over *anything* in the bottom half of the frame.

5. Related Work

VLA foundation models and open-vocabulary policies. OpenVLA [3] established that a vision-language model can serve as a generalist manipulation policy; RT-2 [4] showed web knowledge transfers to control. On the reasoning axis, ACoT-VLA [1] (AgiBot, CVPR 2026) reframes the semantic-kinematic gap by making the policy deliberate directly in the action space, with an explicit reference-trajectory reasoner (EAR) and an implicit action prior from the VLM's internal representations (IAR); on the scale axis, the AgiBot World platform [2] and its GO-1 foundation model (ViLLA, documented in [2]) demonstrate that massive real-robot data plus a latent-action planner yields strong generalisation. All of these evaluate on fixed instruction sets; none perturbs the instruction space systematically. Our spectrum study is complementary: it measures the *boundary* of the generalisation these systems claim — the grounding reliability on which an action-space reasoner depends.

Language robustness for instruction following. In NLP, instruction perturbation analysis is mature — paraphrase sensitivity [4], adversarial instruction attacks [5], and prompt robustness [6] are standard concerns. The robotic literature has borrowed the evaluation protocol but not the perturbation taxonomy: physical-robot studies report aggregate success under manually varied phrasings [7]. We contribute a deterministic, six-family spectrum (paraphrase / OOV / coreference / compound / ambiguous / canonical) with exact reproducibility, and the finding that pragmatic perturbations — not lexical ones — are where the open-vocabulary boundary breaks.

Saliency and prior biases in manipulation policies. Prior work has observed that policies inherit object biases from their training data (e.g., colour preferences in tabletop manipulation [8]). We make the bias measurable: under unconstrained instructions the fallback is non-uniform in every one of four colour-permuted layouts ($\chi^2 p \leq 7.9 \times 10^{-5}$ each), the anchor colour moves with the layout,

and in the original layout the anchor is a deterministic function of the surface phrasing (Fisher exact $p = 7.9 \times 10^{-6}$). What was a qualitative anecdote becomes a statistically testable, and therefore correctable, prior.

6. Reproducibility

- **Model:** `Qwen/Qwen2.5-VL-3B-Instruct`, bf16, offline
- **Environment:** custom `MultiLift` (robosuite 1.4.1 / MuJoCo 2.3.7), deterministic per seed, agentview camera @ 256×256
- **Execution:** CPU software rendering (llvmpipe); CUDA for the VLM
- **Full sweep:** 480 cells (6 families × 4 colours × 5 variants × 4 layouts)
- **Data:** `data/sweep/sweep.json`, `data/sweep/summary.json`, `data/sweep_v1..v3/sweep.json`; **CROSS-**layout claims re-derived by `scripts/cross_scene_facts.py`

References

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